NAME:THOTA LOKESH

REG.NO:192312196

Ex 14: Alpha-Beta Pruning

CODE:

```
def alpha_beta(depth, node_index, is_max, values, alpha, beta, max_depth):
  if depth == max_depth:
    return values[node_index]


  if is_max:
    best = -float('inf')
    for i in range(2):
      val = alpha_beta(depth + 1, node_index * 2 + i, False, values, alpha, beta, max_depth)
      best = max(best, val)
      alpha = max(alpha, best)
      if beta <= alpha:
        break
    return best
  else:
    best = float('inf')
    for i in range(2):
      val = alpha_beta(depth + 1, node_index * 2 + i, True, values, alpha, beta, max_depth)
      best = min(best, val)
      beta = min(beta, best)
      if beta <= alpha:
        break
    return best
```

leaves = [3, 5, 6, 9, 1, 2, 0, -1]

print("Optimal score with pruning:", alpha_beta(0, 0, True, leaves, -float('inf'), float('inf'), 3))

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-14.py =============
Optimal score with pruning: 5
```